

Mathematics - Probability and Statistics

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Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Basic Probability & Counting Rules

Session 5

Basic Probability & Counting Rules

Session 5

Objectives:

- Understand the basic concepts of probability and how they describe uncertainty.
- Apply the fundamental rules of probability to simple experiments.
- Connect probability principles to real-life problems such as dice, cards, lotteries, and passwords.

Probability vs statistics

Probability vs statistics

Probability:

The doctrine of chances. Tells you how often different kinds of events will happen.

- What are the chances of a fair coin coming up heads 10 times in a row?
- If I roll two six sided dice, how likely is it that I'll roll two sixes?

probabilistic questions start with a known model of the world, and we use that model to do some calculations.

the *model* is known, but the *data* are not.

Statistics:

We *do not* know the truth about the world. All we have is the data, and it is from the data that we want to *learn* the truth about the world.

- If my friend flips a coin 10 times and gets 10 heads, are they playing a trick on me?

work out which “model of the world” I should put my trust in

We have data and the *model* is unknown.

Basic Language of Probability

- An experiment that can have different results, even though it is repeated in the same manner every time, is called a **random experiment**.
- Each individual result that is possible for a random experiment is an **outcome**
- The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted as **S, Ω**

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 - Ex: If the outcome of an experiment consists in the determination of the sex of a newborn child, then $\Omega = \{g, b\}$
 - If we toss a coin twice then $\Omega = \{HH, HT, TH, TT\}$

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- An **event is a subset** of the sample space of a random experiment.
 - Ex: if $E = \{g\}$, then E is the event that the child is a girl
 - The event that the first toss is heads is $A = \{HH, HT\}$

Basic Language of Probability

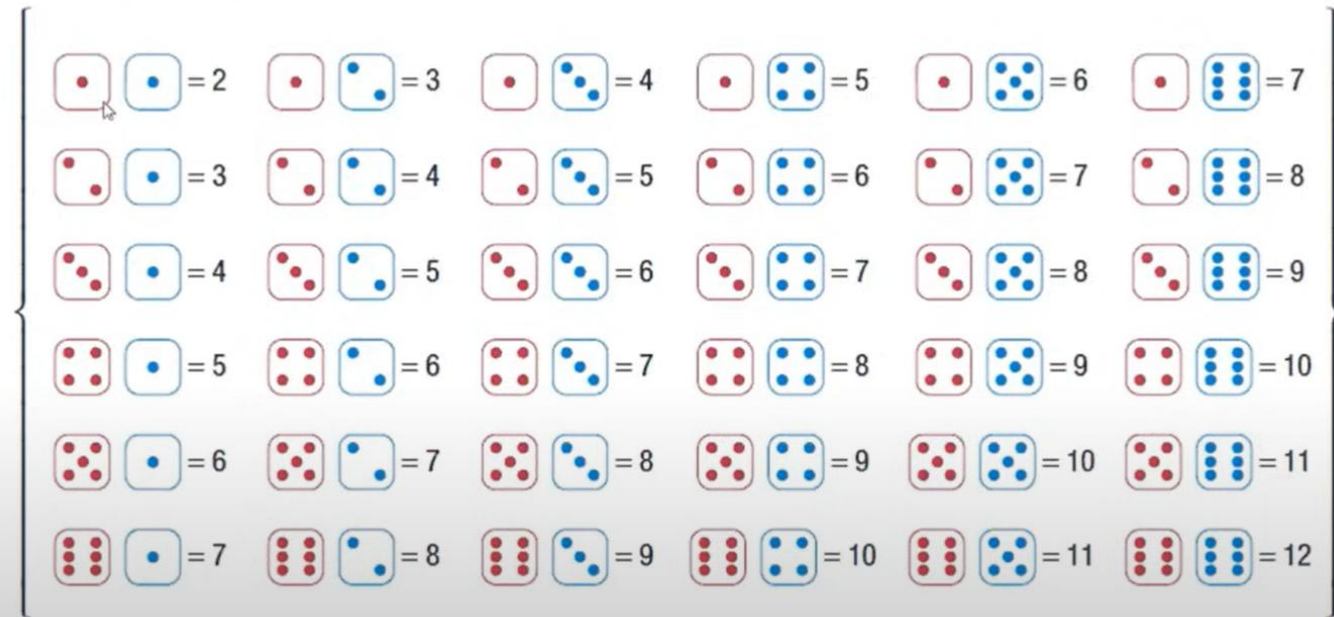
Example: Consider the experiment in which a red six-sided die and a blue six-sided die are rolled together.

- 1) List all the outcomes in the sample space
- 2) List the outcomes in the event « the sum of the numbers rolled on the two dices is equals 6

Basic Language of Probability

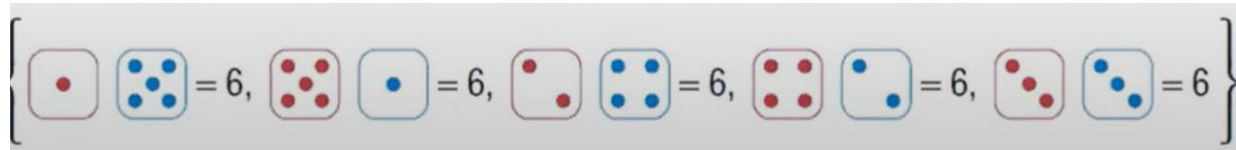
1) Sample space

Sample space =



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2) =6

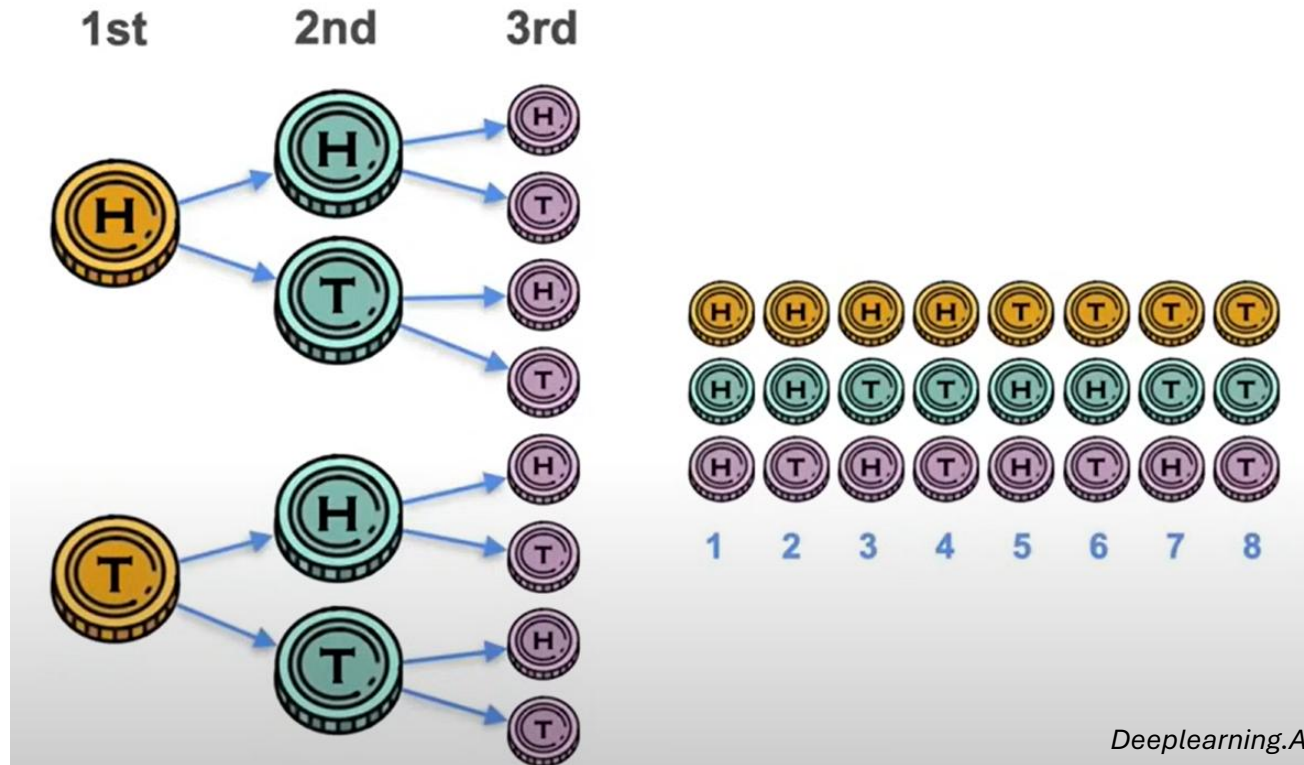


Tree diagram

- A tree diagram is a way to represent outcomes when an experiment has several stages.
- The diagram starts with branches for the first stage outcomes, then extends with branches for each additional stage.
- Each path from the start to the end of the tree represents one unique outcome in the sample space.
- The number of endpoints in the final row equals the total number of possible outcomes.

Tree diagram

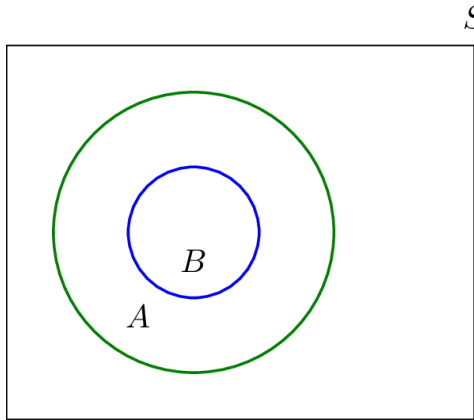
Example: 3 coin tosses



Events: Union, Intersection, Disjoint

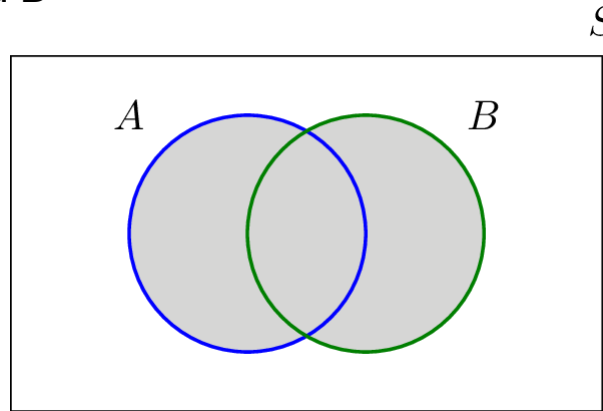
Sample Space S :

We have two events A and B



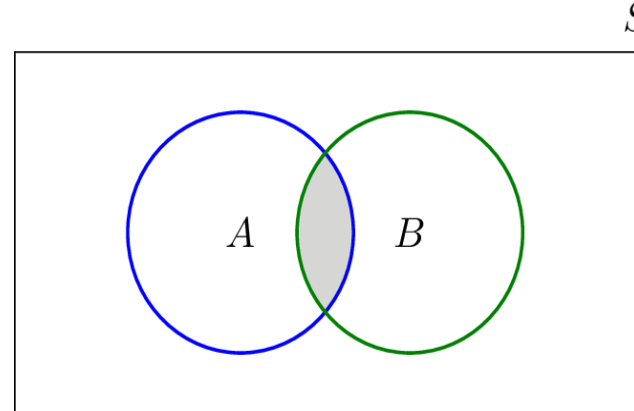
B is included in A

$$B \subset A$$

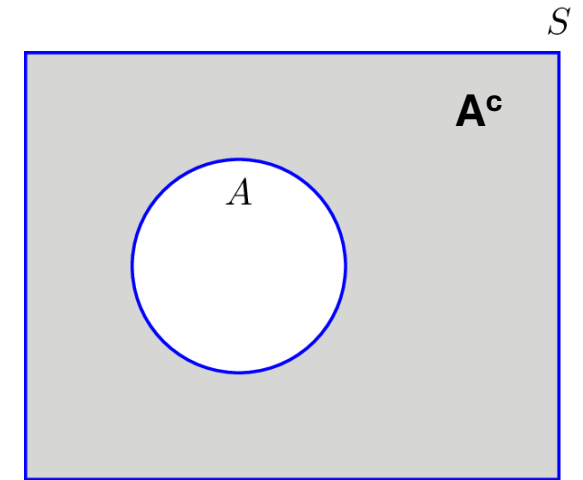


The **union** of two sets is a set containing all elements that are **in A or in B**

$$A \cup B = B \cup A$$



The **intersection** of two events is the event that consists of all outcomes that are contained in **both of the two events**
 $B \cap A = A \cap B$



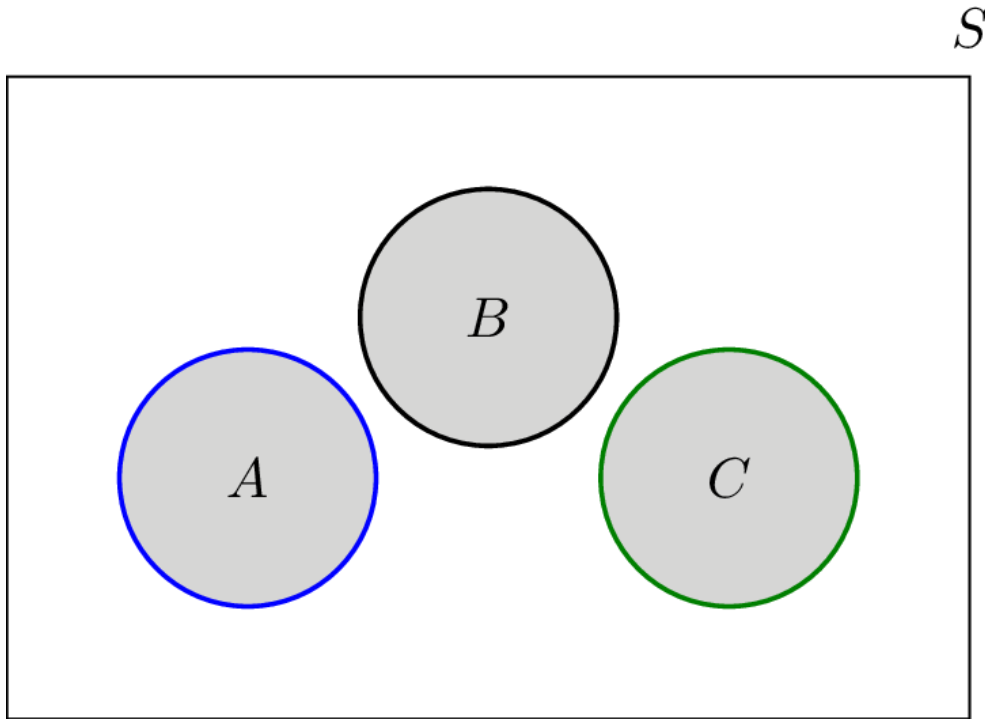
The **complement** of an event in a sample space is the set of **outcomes in the sample space that are not in the event**
Notation: A^c
 $\Omega = A \cup A^c$

A **Venn diagram** is a widely used diagram style that shows the logical relation between sets

Events: Union, Intersection, Disjoint

Sample Space S :

We have two events A and B



Two sets A and B
are **disjoint** if they do not have any
shared element:

$$A \cap B = \emptyset$$

Events: Union, Intersection, Disjoint

Rules:

- The distribution law for set operations implies that
 $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ and $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
- DeMorgan's laws imply that
 $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$
- Also as mentioned previously:
 $A \cup B = B \cup A$ and $A \cap B = B \cap A$

Example:

If the universal set is given by $S = \{1, 2, 3, 4, 5, 6\}$, and $A = \{1, 2\}$, $B = \{2, 4, 5\}$, $C = \{1, 5, 6\}$ are three sets, find the following sets:

- $A \cup B$
- $A \cap B$
- A^c
- B^c
- Check De Morgan's law by finding $(A \cup B)^c$ and $A^c \cap B^c$.
- Check the distributive law by finding $A \cap (B \cup C)$ and $(A \cap B) \cup (A \cap C)$

Events: Union, Intersection, Disjoint

Example:

If the universal set is given by $S=\{1,2,3,4,5,6\}$, and $A=\{1,2\}$, $B=\{2,4,5\}$, $C=\{1,5,6\}$ are three sets, find the following sets:

- $A \cup B: \{1,2,4,5\}$
- $A \cap B: \{2\}$
- $A^c: \{3,4,5,6\}$
- $B^c: \{1,3,6\}$
- Check De Morgan's law by finding $(A \cup B)^c$ and $A^c \cap B^c$:
 - $A^c \cap B^c = \{3,4,5,6\} \cap \{1,3,6\} = \{3,6\}$
 - $(A \cup B)^c = \{1,2,4,5\}^c = \{3,6\}$
- Check the distributive law by finding $A \cap (B \cup C)$ and $(A \cap B) \cup (A \cap C)$
 - $A \cap (B \cup C) = \{1,2\} \cap (\{2,4,5\} \cup \{1,5,6\}) = \{1,2\} \cap \{1,2,4,5,6\} = \{1,2\}$
 - $(A \cap B) \cup (A \cap C) = (\{1,2\} \cap \{2,4,5\}) \cup (\{1,2\} \cap \{1,5,6\}) = \{2\} \cup \{1\} = \{1,2\}$

Probability and axioms

Exercise:

The following table summarizes visits to emergency departments at four hospitals in Arizona. People may leave without being seen by a physician, and those visits are denoted as LWBS. The remaining visits are serviced at the emergency department, and the visitor may or may not be admitted for a stay in the hospital.

let A denote the event that a visit is to Hospital 1 and let B denote the event that the result of the visit is LWBS.

Calculate the number of outcomes in

- $A \cap B$
- $A \cup B$
- A^c

Result	Hosp 1	Hosp 2	Hosp 3	Hosp 4	Total
LWBS	195	270	246	242	953
Admitted	1277	1558	666	984	4485
Not adm.	3820	5163	4728	3103	16,811
Total	5292	6991	5640	4329	22,252

Probability and axioms

Exercise:

A denote the event that a visit is to Hospital 1

B denote the event that the result of the visit is LWBS

- $A \cap B$: 195
- $A \cup B$: $5292 + 270 + 246 + 242 = 6050$
- $A^c = 22,252 - 5292 = 16,960$

Result	Hosp 1	Hosp 2	Hosp 3	Hosp 4	Total
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Probability and axioms

Definition:

The probability that an event E occurs is given by:

$$P(E) = \frac{n(E)}{n(S)}$$

where

- $n(E)$ is the number of outcomes in the event.
- $n(s)$ is the number of outcomes in the sample space.

Axioms:

- **Non negativity:** $0 \leq P(A) \leq 1$
- **Normalization:** $P(\Omega) = 1$
- **Additivity:** If A_i disjoint $P(\cup A_i) = \sum P(A_i)$
- **Inclusion:** If $B \subset A$ therefore $P(B) \leq P(A)$ (e.g face vs king)

Therefore:

- $P(\emptyset) = 0$
- $P(A^c) = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (A and B disjoint events, $P(A \cup B) = P(A) + P(B)$)

Probability and axioms

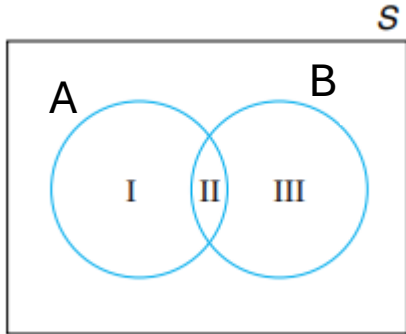
Exercise: Demonstrate

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Probability and axioms

Exercise:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$P(A \cup B) = P(I) + P(II) + P(III)$$

$$P(A) = P(I) + P(II)$$

$$P(B) = P(III) + P(II)$$

$$P(A \cap B) = P(II)$$

$$\begin{aligned} \text{Therefore: } P(A) + P(B) - P(A \cap B) &= P(I) + P(III) + 2P(II) - P(II) \\ &= P(A \cup B) \end{aligned}$$

Probability and axioms

Exercise:

A total of 28 percent of males living in Nevada smoke cigarettes, 6 percent smoke cigars, and 3 percent smoke both cigars and cigarettes. What percentage of males smoke neither cigars nor cigarettes?

Probability and axioms

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A total of 28 percent of males living in Nevada smoke cigarettes, 6 percent smoke cigars, and 3 percent smoke both cigars and cigarettes. What percentage of males smoke neither cigars nor cigarettes?

Solution:

A= event that the male is a cigarettes smoker

B= event that the male is a cigar smoker

Probability that the male is a cigar or cigarette smokers is:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.28 + 0.06 - 0.03 = 0.31 \text{ (31\%)}$$

Therefore, the probability the male is not a smoker is: $1 - P(A \cup B) = 1 - 0.31 = 0.69$ (69%)

Probability and axioms

Exercise:

What is the probability of obtaining a sum of 7 or a sum of 10 with 2 dices?

Probability and axioms

Exercise:

What is the probability of obtaining a sum of 7 or a sum of 10 with 2 dices?

Solution:

- $n(A)=6$
- $n(B)=3$

So number of event possible: 9

$$n(\Omega)=36$$

$$\begin{aligned} P(\text{Sum of 7 or sum of 10}) &= (n(A)+n(B))/n(\Omega) \\ &= 9/36 \\ &= 1/4 \text{ (25\%)} \end{aligned}$$

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

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Probability and axioms

Exercise:

What is the probability of obtaining a sum of 7 or a difference of one with two dices?

Probability and axioms

Exercise:

What is the probability of obtaining a sum of 7 or a difference of one with two dices?

Solution:

- $n(A)=6$
- $n(B)=10$
- $n(A \cap B)=2$
- $n(\Omega)=36$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 6/36 + 10/36 - 2/36 \\ &= 14/36 \\ &= 7/18 \end{aligned}$$

sum = 7



1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

diff = 1



1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

Probability and axioms

Exercise:

At school kids can play only 1 sport: football or rugby

The probability that a kid play football is $P(A)=0.3$

The probability that a kid play rugby is $P(B)= 0.4$

What is the probability that a kid play football or rugby.

Probability and axioms

Exercise:

At school kids can play only 1 sport: football or rugby

The probability that a kid play football is $P(A)=0.3$

The probability that a kid play rugby is $P(B)= 0.4$

What is the probability that a kid play football or rugby.

Solution:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.4 - 0 \text{ (Kids can play **only ONE** sport!!)} \\ = 0.7 \text{ (70\%)}$$

Multiplication rules

- A multistage experiment is an experiment with two or more steps/stages
- An experiment performed **with replacement** refers to placing objects back into consideration before performing the next step of the experiment.
- Two events **are independent** if one event happening **does not influence the probability of the other event** happening

Counting techniques

Multiplication technique:

- Assume an operation can be described as a sequence of k steps
- The number of ways of completing step 1 is n_1 ,
- the number of ways of completing step 2 is n_2 for each way of completing step 1
- the number of ways of completing step 3 is n_3 for each way of completing step 2, and so forth.

The total number of ways of completing the operation is $n_1 \times n_2 \times \dots \times n_k$

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The total number of ways of completing the operation is $n_1 \times n_2 \times \dots \times n_k$

Exemple: How many outcomes you have if you toss 3 times a coin?

1st: 2 outcomes (H or F)

2nd: 2 outcomes (H or F)

3rd: 2 outcomes (H or F)

Total $2^3=8$

Counting techniques

Exercise: A telephone number is a 7-digit sequence, but the first digit has to be different from 0 or 1. How many distinct telephone numbers are there?

Counting techniques

Sol: A telephone number is a 7-digit sequence, but the first digit has to be different from 0 or 1. How many distinct telephone numbers are there?

First digit, possibility: 2,3,4,5,6,7,8,9= 8 outcomes

Second digit: 0,1,2,3,4,5,6,7,8,9 =10 outcomes

....

Number total of distinct numbers: $8 \cdot 10^6$

Counting techniques

Exercise: How many possible license plates could be stamped if each license plate were required to have exactly 3 letters and 4 numbers?

Counting techniques

Sol: How many possible license plates could be stamped if each license plate were required to have exactly 3 letters and 4 numbers?

1/26 (letters: a,b,c,d,e,f,g,h,i,j,k,l,m,n,o,p,q,r,s,t,u,v,w,x,y,z)

1/10 (0,1,2,3,4,5,6,7,8,9)

$$26*26*26*10*10*10*10=\mathbf{175,650,000}$$

Factorial

A factorial is the product of all positive integers less than or equal to a given positive integer, n . Therefore, it is given by:

$$n! = n(n-1)(n-2)\dots(2)(1)$$

- **$0! = 1$**

Examples:

- $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$
- $5! / (5-3)! = 5! / 2! = 5 \times 4 \times 3 \times 2 \times 1 / (2 \times 1) = 5 \times 4 \times 2 = 60$
- $45! / 43! = (45 \times 44 \times 43 \times 42 \times \dots \times 2 \times 1) / (43 \times 42 \times \dots \times 2 \times 1) = 45 \times 44 = 1980$

Generalization of the Multiplication Principle

Suppose there are n positions to be filled with n different objects, in which there are:

- n choices for the 1st position
- $(n-1)$ choices for the 2nd position
- $(n-2)$ choices for the 3rd position
- ... and ...
- 1 choice for the last position

The Multiplication Principle tells us there are then in general:

$$n(n-1)(n-2)\cdots 1=n!$$

Permutations

For example you want to derive the number of **ordered sequences** that can be formed from the elements of a set: $S=\{a,b,c\}$, so here $n=3$

From the formula you should have $3!$ different sequences. Indeed, we have : *abc, acb, bac, bca, cab, and cba*

A **permutation** of S is any arrangement of its elements in a specific order.

ORDER MATTERS

Example: How many ways can 4 people fill 3 chairs?

First chair: 4 options, Second chair: 3, third chair: 2 $\rightarrow 4 \cdot 3 \cdot 2 = 24$

The key difference between this example and the previous one is that the first involved arranging all 4 people, while this one leaves out one person and arranges only 3 of the 4. This setup leads us to a useful generalization of the Multiplication Principle: the number of **permutations of n objects taken r at a time where $k \leq n$**

Permutations

The number of permutations of subsets of k elements selected from a set of n different elements is

$$P(n, k) = \frac{n!}{(n - k)!}$$

If you define: $\binom{n}{k} = \frac{n!}{k!(n - k)!}$, therefore $P(n, k) = \binom{n}{k} \cdot k!$

Ex: How many ways can 4 people fill 3 chairs?

$$P(4, 3) = 4! / (4 - 3)! = 4! / 1! = 4 \cdot 3 \cdot 2 \cdot 1 / 1 = 4 \cdot 3 \cdot 2 = 24$$

Combinations

Another counting problem of interest is the number of subsets of k elements that can be selected from a set of n elements. Here, order is not important. These are called **combinations**

ORDER DOES NOT MATTER

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n - k)!}$$

Example: You have a group of 10 students, you have to pick 3 to make a group. How many combinations do you have?

$$C(10, 3) = 10! / (3!(10 - 3)!) = 10! / 3!7! = 10 \cdot 9 \cdot 8 / 6 = 120$$

Special permutation

Until now, permutations assumed **all objects are distinct**. But what if some are identical? Then many of the permutations look the same, so we are overcounting.

The number of distinguishable permutations of n objects, of which n_1 are all alike, of which n_2 are all alike, Is given by:

$$\frac{n!}{n_1!n_2!\dots n_k!}$$

Example: The word **BANANA** has 6 letters. If all letters were unique, we'd have $6!=720$ permutations. But because letters repeat (3 A's and 2 N's), many permutations are duplicates. The corrected count is

$$\frac{6!}{3! \cdot 2!} = 60$$

Exercices

Now the ship's captain sends signals by arranging 3 red, 4 orange and 2 blue flags on a vertical pole. How many different flag signals could the ship's captain possibly send?

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Now the ship's captain sends signals by arranging 3 red, 4 orange and 2 blue flags on a vertical pole. How many different flag signals could the ship's captain possibly send?

Solution:

RRR
OOOO
BB

$$\frac{n!}{n_1!n_2!\dots n_k!}$$

There is 9 flags, so 9! Possibilities. However, red flags are the same (as blue, and orange) so we need to take into account this.

3! for R
4! for orange
2! for blue

9!/(4!3!2!)=1260

Exercices

Given a group of three friends, Paul, Pierre, Jacques:

- How many ways can you arrange the way they stand in line?

Sol: So stand in line, order matters. $P(n, k) = \frac{n!}{(n - k)!}$ so: $3! / (3-3)! = 3! = 6$

How many ways can you select two of them?

Sol: Order does not matter:

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n - k)!} = 3! / 2!(3-2)! = 3 \text{ (Paul/Pierre, Paul/Jacques, Pierre/Jacques)}$$

Exercices

You have a class of 18 students. They want to vote for class president, vice president, and secretary.

How many different ways can the officers be elected?

Exercices

You have a class of 18 students. They want to vote for class president, vice president, and secretary.

How many different ways can the officers be elected?

Sol: The order of the students chosen is important therefore it is a permutation problem, with $n=18$ and $k=3$

$$P(n, k) = \frac{n!}{(n - k)!} \quad \text{So: } 18!/15!=18*17*16=4896$$

Exercices

Maya has a bag with 15 balls, each of which is a different color including red, blue, and yellow. Maya pulls out 3 balls. What is the **probability** that the blocks she has chosen are red, blue, and yellow?

Exercices

Maya has a bag with 15 balls, each of which is **a different color** including red, blue, and yellow. Maya pulls out 3 balls. What is the **probability** that the blocks she has chosen are read, blue, and yellow?

Sol: Order does not matter

First, what is the number of outcomes in the sample space?

$$C(15,3) = \frac{15!}{3!(12!)} = \frac{15 \cdot 14 \cdot 13}{3 \cdot 2 \cdot 1} = 5 \cdot 7 \cdot 13 = 455$$
$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Second, how many combinations contain the red, blue, and yellow balls?

There is **only 1 combination** that satisfies this (since each ball has a unique color).

Therefore, $P(\text{red, blue, yellow}) = 1/455$

Exercices

A group of 12 tourists is visiting le Louvre. There is a discounted admission for group of at least ten.

How many combinations of tourists can be made for le Louvre so that the group receives the discounted rate?

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A group of 12 tourists is visiting le Louvre. There is a discounted admission for group of at least ten.

How many combinations of tourists can be made for le Louvre so that the group receives the discounted rate?

Sol: Order does not matter. If at least 10 are required then the group will get discount if 10, 11, or 12 tourists go to le Louvre. We must calculate the number of combinations for each of the 3 possibilities

$C(12,10)=66$, $C(12,11)=12$, $C(12,12)=1$, therefore, the total combination is: 79

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n - k)!}$$

Exercices

What is the probability of having 5 hearts if I pick 5 cards in a 52 card deck

Exercices

What is the probability of having 5 hearts if I pick 5 cards in a 52 card deck

Sol: Order does not matter -> combination

$$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n - k)!}$$

- Number of way to take 5 cards?

$$C(52, 5) = 52! / 5!(52 - 5)! = 52 * 51 * 50 * 49 * 48 / (5 * 4 * 3 * 2 * 1) = 2,598,960$$

- Number of way to have all hearts

$$C(13, 5) = 13! / 5!(13 - 5)! = (13 * 12 * 11 * 10 * 9) / (5 * 4 * 3 * 2 * 1) = 1287$$

$$\text{Therefore, } P(\text{having 5 hearts}) = 1287 / 2598960 = 0.00049 = 0.049\%$$

Exercices

What is the probability of drawing 5 Hearts in a Specific Order from a 52 card-deck?
(like A♥-2♥-3♥-4♥-5♥, or 5♥-6♥-7♥-8♥-9♥)

Exercices

What is the probability of drawing 5 Hearts in a Specific Order from a 52 card-deck?
(like A♥-2♥-3♥-4♥-5♥, or 5♥-6♥-7♥-8♥-9♥)

Sol: Order matters -> permutation

$$P(n, k) = \frac{n!}{(n - k)!}$$

- Number of way to take 5 cards but without replacement and in order?

$$P(52, 5) = 52! / (52 - 5)! = 52! / 47! = 52 * 51 * 50 * 49 * 48 = 311,875,200$$

- Number of way to drawing 5 Hearts in a Specific Order

10 possibilities: A2345, 23456, 34567, 45678, 56789, 678910, 78910J, 8910JQ, 910JQK, 10JQKA

For each possibility, you have $5! / (5 - 5)! = 5!$ possibilities ($\{5♥, 6♥, 7♥, 8♥, 9♥\}$: 9♥, 8♥, 7♥, 6♥, 5♥, or 7♥, 5♥, 9♥, 8♥, 6♥}, etc)

Therefore, $P(\text{having 5 hearts in a sequence}) = 5! * 10 / 311,875,20 = 0.000038 = 0.0038\%$